

Reminders

Upcoming due dates

Wed Assignment 2

Look for your Proposal feedback tonight/
tomorrow!

Fri Discussion lab 5

Dimensionality reduction

Data Science in Practice

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Dimensionality Reduction

A mathematical process to reduce the number of variables to consider



Discuss: Why? What are the use cases?

Two ways to use DR

- Like EDA for wide datasets
 - Gain insights
 - Understand how different variables relate to one another
- As a feature transformation to input into a model or predictor
 - Smaller # of variables means less data needed and easier learning or modeling

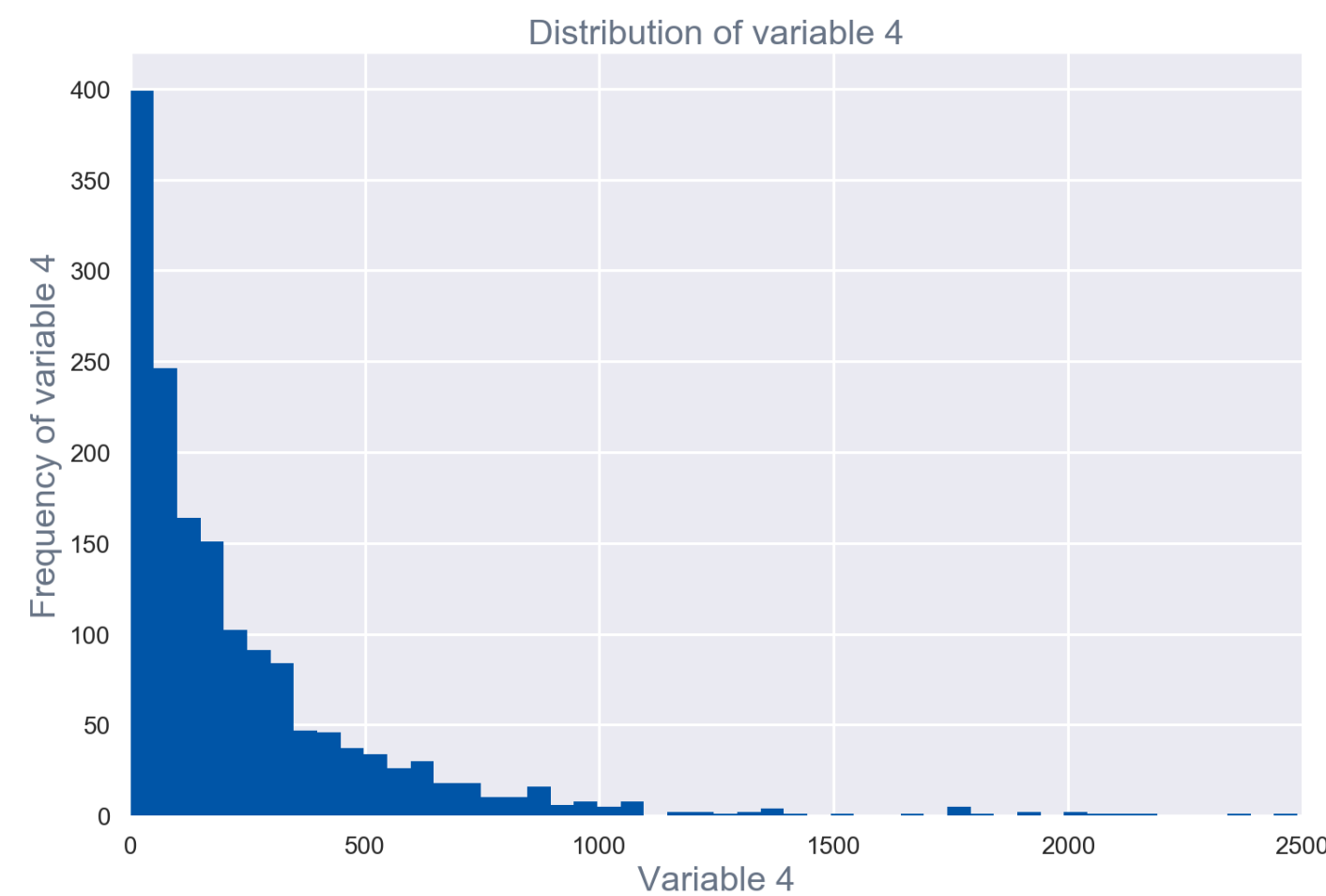
Use Cases for Dimensionality Reduction

- Thousands of sensors used to monitor an industrial process
 - Reducing the data from these 1000s of sensors to a few features, we can then build an interpretable model
 - Goal : predict process failure from sensors
- Understanding diet around the world
 - Amount of foods eaten among populations across the world
 - Goal: identify diet similarity among populations
- Identify genetic ancestry
 - Determine ancestral origins based on genetic variation
 - Goal: Learn more about our genetic history

Exploratory

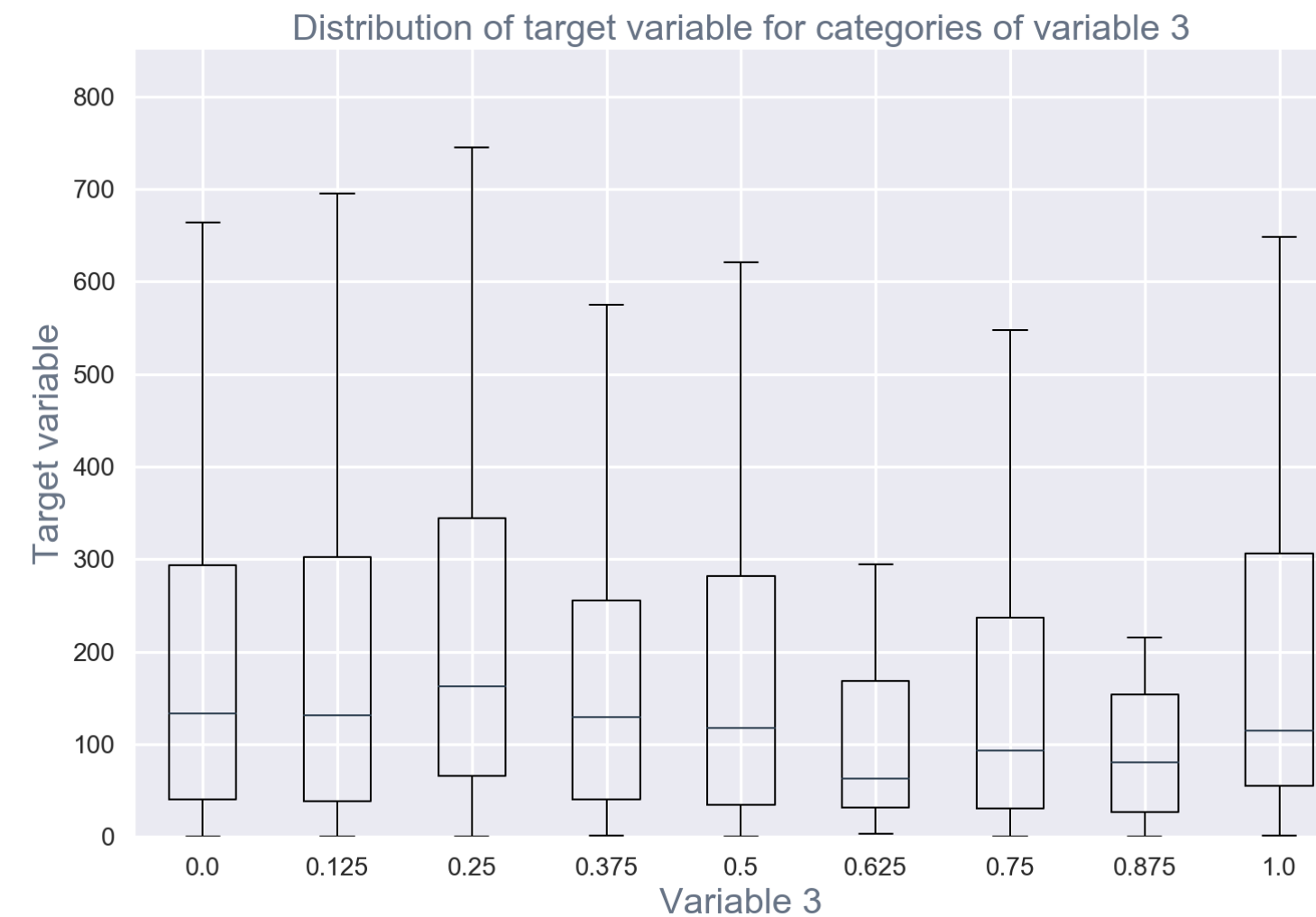
EDA Approaches to “Get a Feel for the Data”

Understanding the relationship between variables in your dataset



Univariate

understanding a single variable
i.e.: histogram, densityplot, barplot



Bivariate

understanding relationship between 2 variables
i.e.: boxplot, scatterplot, grouped barplot, boxplot



Dimensionality Reduction

projecting high-D data into a lower-D space
i.e.: PCA, ICA, Clustering

Principal Component Analysis (PCA)

Key Terms:

- **Principal Component (PC)** - a linear combination of the predictor variables
- **Loadings** - the weights that transform the predictors into components (aka weights)
- **Screeplot** - Variance explained of each component

Principal Component Analysis (PCA)

Goal : combine multiple numeric predictor variables into a smaller set of variables. Each variable in this smaller set is a weighted linear combination of the original set.

This smaller set of variables -- the *principal components* (PCs) - “explain” most of the variability of the full set of variables....but uses many fewer dimensions to do so.

The weights (loadings) used to form the PCs explain the relative contributions of the original variables to the new PCs.

PCA

$$\forall i \in [1, n] : \quad Z_i = \sum_{\forall j \in [1, n]} w_{ij} X_j$$

X_j are in the original input space, Z_i are in the PC output space

w_{ij} are weights (aka *loadings*) that transform the original variables into principal components

Z_1 : the first principal component (PC1) which is the linear combination that best explains the total variance

For a simple 2D inputs, 2D outputs PCA the equations look like

$$Z_1 = w_{1,1}X_1 + w_{1,2}X_2$$

$$Z_2 = w_{2,1}X_1 + w_{2,2}X_2$$

Stock Price returns for Chevron (CVX) and ExxonMobil (XOM)

PC1 and PC2 are the dotted lines on the plot

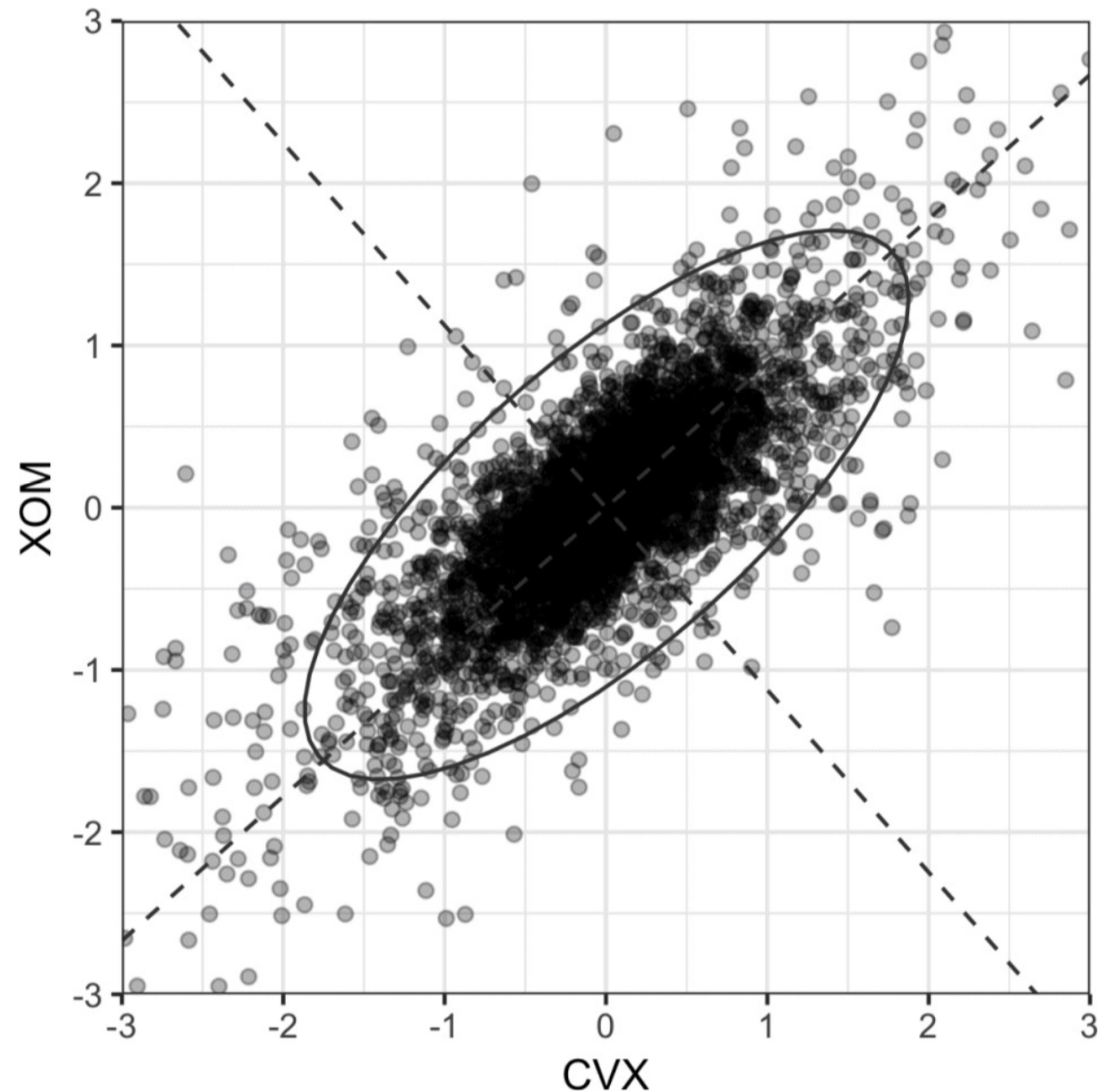


Figure 7-1. The principal components for the stock returns for Chevron and ExxonMobil

PCA

Eigenvectors

Eigenvalues

$$A\vec{v} = \lambda\vec{v}$$

1. Center the data
2. Compute covariance matrix
3. Compute eigenvectors and eigenvalues of covariance matrix
4. (Optional) subset eigenvectors by picking the m largest eigenvalues
5. Project datapoints into PCs by multiplying with eigenvectors

PCA doesn't work if the data isn't zero mean!

Typically we use a z-score if data is close to normally distributed

Principal Component Analysis (PCA)

But....PCA shines when you're dealing with high-dimensional data. So we have to move *beyond* two predictors to many predictors....

Step 1: Combine all predictors in linear combination

Step 2: Assign weights that optimize the collection of the covariation to the first PC (Z_1) (maximizes the % total variance explained)

Step 3: Repeat Step 2 to generate new predictor Z_2 (second PC) with different weights. By definition Z_1 and Z_2 are uncorrelated. Continue until you have as many new variables (PCs) as original predictors

Step 4: Retain as many components as are needed to account for *most* of the variance.

S&P 500 Data: 5648 days (1993-2015) x 517 stocks

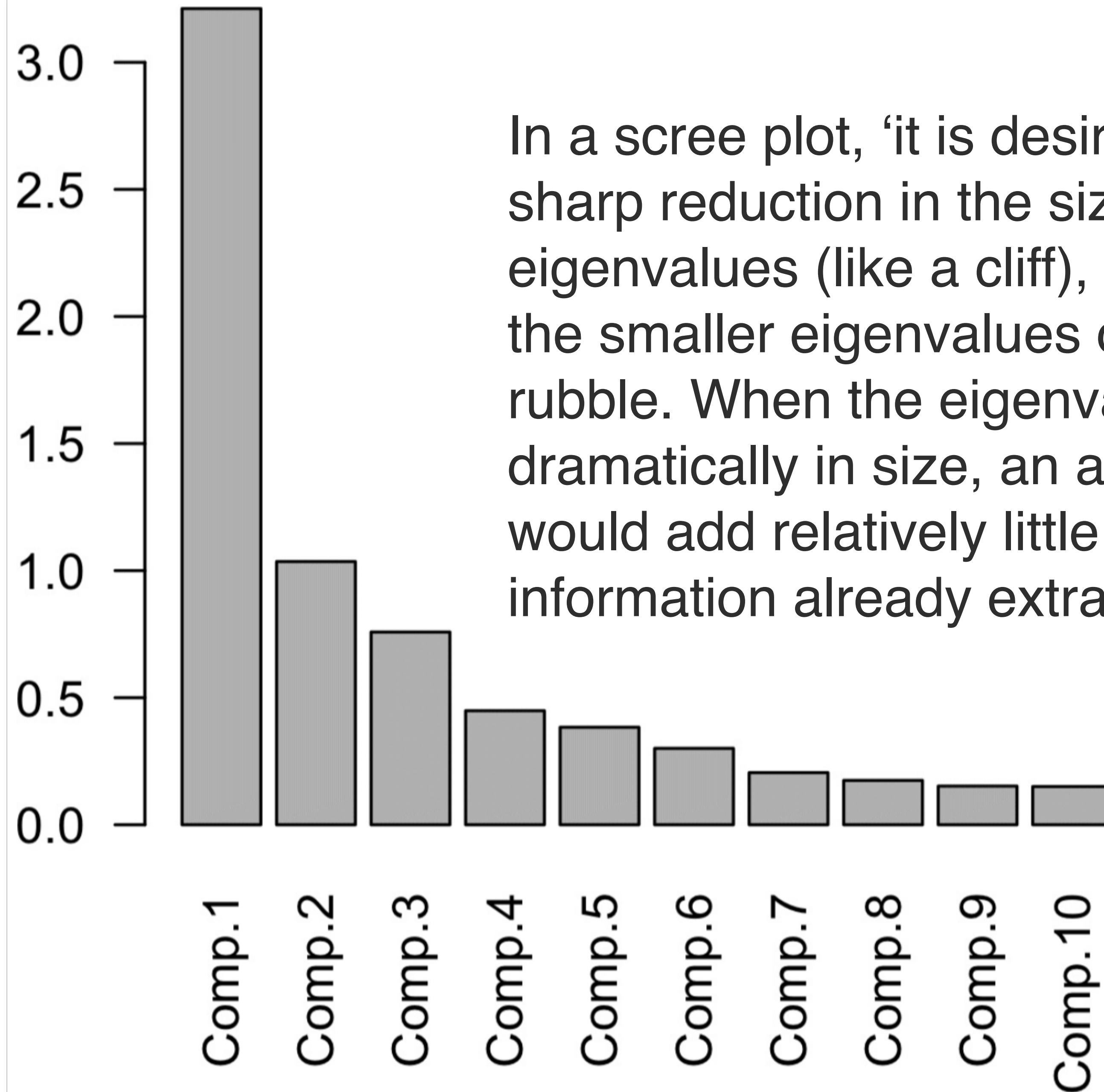
	ADS	CA	MSFT	RHT	CTSH	CSC	EMC	IBM	XRX	ALTR	ADI	AVGO	BRCM	FSLR	INTC	LLTC	MCHP	MU	NVDA
1/29/93	0	0.06012444	-0.0220998	0	0	0.01889746	0.00736807	0.0921652	0.25914009	-0.0071053	-0.0157849	0	0	0	-0.0504878	-0.0898696	0	0.03702057	0
2/1/93	0	-0.180389	0.02762115	0	0	0.01888884	0.01842489	0.11520651	-0.1007745	0.06389288	-0.0157929	0	0	0	0.09536733	0.0449348	0	0.03702038	0
2/2/93	0	-0.1202566	0.03589987	0	0	-0.0755726	0.02948172	-0.0230413	0.02879553	-0.0141924	0.0473628	0	0	0	0	0.0674022	0	0.12340155	0
2/3/93	0	0.0601242	-0.024857	0	0	-0.151128	0.00368875	-0.2534543	-0.04319	-0.0071053	0.20523612	0	0	0	-0.050495	0.0224674	0	-0.0123403	0
2/4/93	0	-0.3607697	-0.0607567	0	0	0.11335029	-0.0221136	0.0698618	0	-0.0070962	-0.0315699	0	0	0	0	0.0224674	0	-0.0740409	0
2/5/93	0	0.03005777	0.09389247	0	0	0.09445283	-0.0479066	0.04657454	0.17276006	-0.0212976	-0.0631478	0	0	0	-0.0476873	-0.0674022	0	-0.0123403	0
2/8/93	0	0.03006643	-0.0607498	0	0	-0.1133503	-0.0110568	0.11643635	-0.04319	0.00709618	0	0	0	0	-0.0196321	-0.1235743	0	-0.0617008	0
2/9/93	0	-0.0901902	-0.063521	0	0	-0.1322391	-0.0147456	0.06986181	-0.115169	0.04969143	-0.0157929	0	0	0	-0.0112235	0.0224674	0	0	0
2/10/93	0	0.12025657	0.02209981	0	0	0.09445283	0.01474557	-0.2561599	0.01439448	0.02838473	0.01578495	0	0	0	0.04487956	0.11233699	0	0.07404095	0
2/11/93	0	0.03005825	-0.0220927	0	0	-0.0188975	0.01474556	-0.1397236	-0.04319	0.02129762	-0.0315699	0	0	0	-0.0532953	0.06740222	0	-0.0246804	0
2/12/93	0	-0.0901901	-0.0358999	0	0	-0.0377863	-0.0073681	-0.0698618	-0.1871546	0	-0.0473628	0	0	0	-0.0336561	-0.112337	0	-0.0370204	0
2/16/93	0	-0.6313411	-0.0607497	0	0	-0.0377863	-0.0479066	-0.0931491	-0.04319	-0.0283938	-0.1262955	0	0	0	-0.098175	-0.1460417	0	-0.0246803	0
2/17/93	0	0.12025657	-0.0165712	0	0	-0.1700254	-0.0110568	0.04657453	-0.08638	-0.0142015	0.03157785	0	0	0	0.04487955	0	0	-0.0123403	0
2/18/93	0	-0.1803808	0.00828562	0	0	-0.0566751	0.00368875	-0.0931491	-0.08638	0	-0.0157849	0	0	0	-0.0168315	0.0224674	0	0.03702056	0
2/19/93	0	0.03006595	-0.0469427	0	0	0	0.00736807	-0.0232873	0.115169	0.01419237	0.03157785	0	0	0	0.10378311	0.15727183	0	0.14808196	0
2/22/93	0	0.03005825	-0.0662782	0	0	-0.1322477	-0.0184249	0.13972361	0	0.02839382	-0.0631557	0	0	0	-0.0168317	-0.0674022	0	0	0
2/23/93	0	-0.0300583	0.03314266	0	0	0	-0.0479066	-0.0698618	-0.1439645	-0.0070962	0.03157785	0	0	0	-0.0336631	-0.0337047	0	-0.0493606	0
2/24/93	0	0.15031459	0.10769942	0	0	0.01888884	0.04421782	0.1397236	0.08638003	0.00709618	0	0	0	0	0.11781411	-0.0224674	0	0.09872137	0
2/25/93	0	0.15032277	0.04142827	0	0	0.01888884	-0.0110568	0.37259628	0	0.02839382	0	0	0	0	0.0112163	0.15727183	0	-0.0370205	0
2/26/93	0	-0.0300659	-0.0193286	0	0	-0.0188888	0.01105682	0.06986181	0.05759106	0	0.01578495	0	0	0	-0.028055	-0.0224674	0	-0.074041	0
3/1/93	0	-0.180381	-0.0497068	0	0	-0.0944614	-0.0073681	0	0.04358505	0.00709618	0.09472561	0	0	0	-0.0336631	0.0224674	0	0	0
3/2/93	0	0	0.06351413	0	0	0.15113659	0.00368875	-0.0698618	0.116229	0	0.11051053	0	0	0	0.09537435	-0.0449348	0	0.12340155	0
3/3/93	0	0.12025658	0	0	0	-0.0566751	0.03684977	0.16301088	0.02905891	-0.0070962	0	0	0	0	0.01402383	-0.112337	0	-0.0370204	0
3/4/93	0	-0.1503146	-0.0220927	0	0	0.0377863	0.00367932	-0.0698618	-0.1452879	-0.0070962	-0.015785	0	0	0	-0.0252473	-0.0674022	0	0.01234016	0
3/5/93	0	0.03005825	-0.0165714	0	0	-0.0944614	0.00368875	-0.0232873	0	0.03549001	0	0	0	0	-0.0617041	0.0449348	0	0.02468042	0
3/8/93	0	0.06012444	0.02209275	0	0	0.01888884	-0.025793	0.11643634	0.21792524	0	0.04736279	0	0	0	0.06731932	0.13480441	0	0.09872114	0
3/9/93	0	0.09019015	0.00552151	0	0	0.09446144	0.00736807	0.09314908	-0.0290523	-0.0070962	-0.0157849	0	0	0	0.0112163	0.0898696	0	0	0
3/10/93	0	0.03006595	0.01104991	0	0	0	0.01105681	-0.1862981	0.02905891	-0.0141924	-0.0157849	0	0	0	-0.0196321	-0.0449348	0	0	0
3/11/93	0	-0.0300583	0.02761408	0	0	0.22670058	0	-0.1862982	-0.0581112	0.00709618	0.06314774	0	0	0	-0.0196392	0.01123011	0	0	0
3/12/93	0	0	0.06627822	0	0	-0.0188975	0.01474556	0.30273448	-0.1452813	0.02839381	0.06314774	0	0	0	0.02524749	0.01123729	0	0.13574153	0

For this example: we'll focus on 16 top companies

Screeplot

The vernacular definition of “scree” is an accumulation of loose stones or rocky debris lying on a slope or at the base of a hill or cliff.

Eigenvalues



In a scree plot, ‘it is desirable to find a sharp reduction in the size of the eigenvalues (like a cliff), with the rest of the smaller eigenvalues constituting rubble. When the eigenvalues drop dramatically in size, an additional factor would add relatively little to the information already extracted.’ ([Source](#))

Figure 7-2. A screeplot for a PCA of top stocks from the S&P 500

How many PCs to select?

Option 1: Visually through the screeplot: elbow method

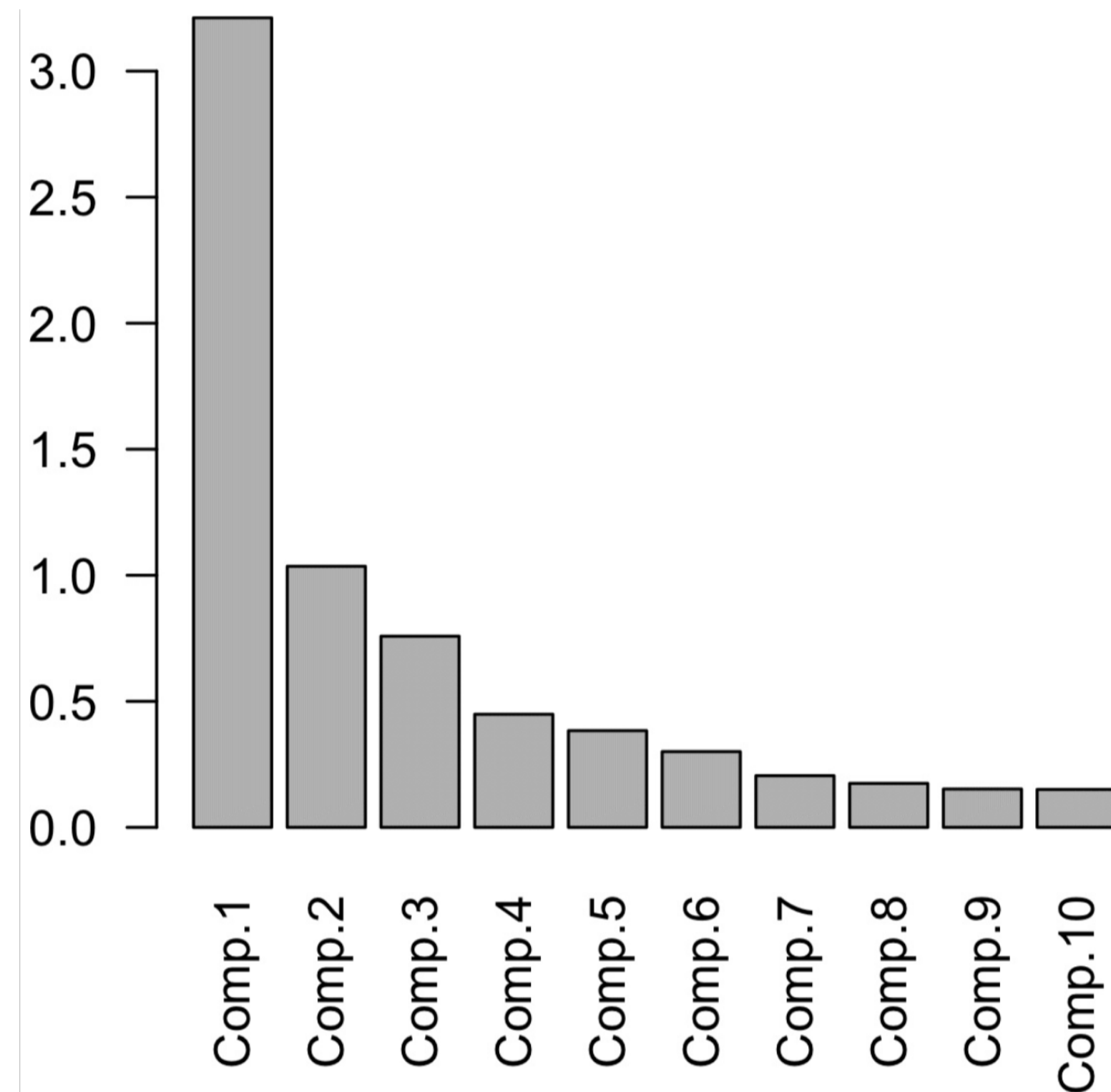


Figure 7-2. A screeplot for a PCA of top stocks from the S&P 500

Option 2: % Variance explained (i.e. 80% variance explained)

Option 3: Inspect loadings for an intuitive interpretation

Option 4: Cross-validation

Loading of PCs 1-5

PC1: Overall stock market trend

PC2: Price change of energy stocks

PC3: movements of Apple and CostCo.

PC4: movements of Schlumberger to other stocks

PC5: Financial companies

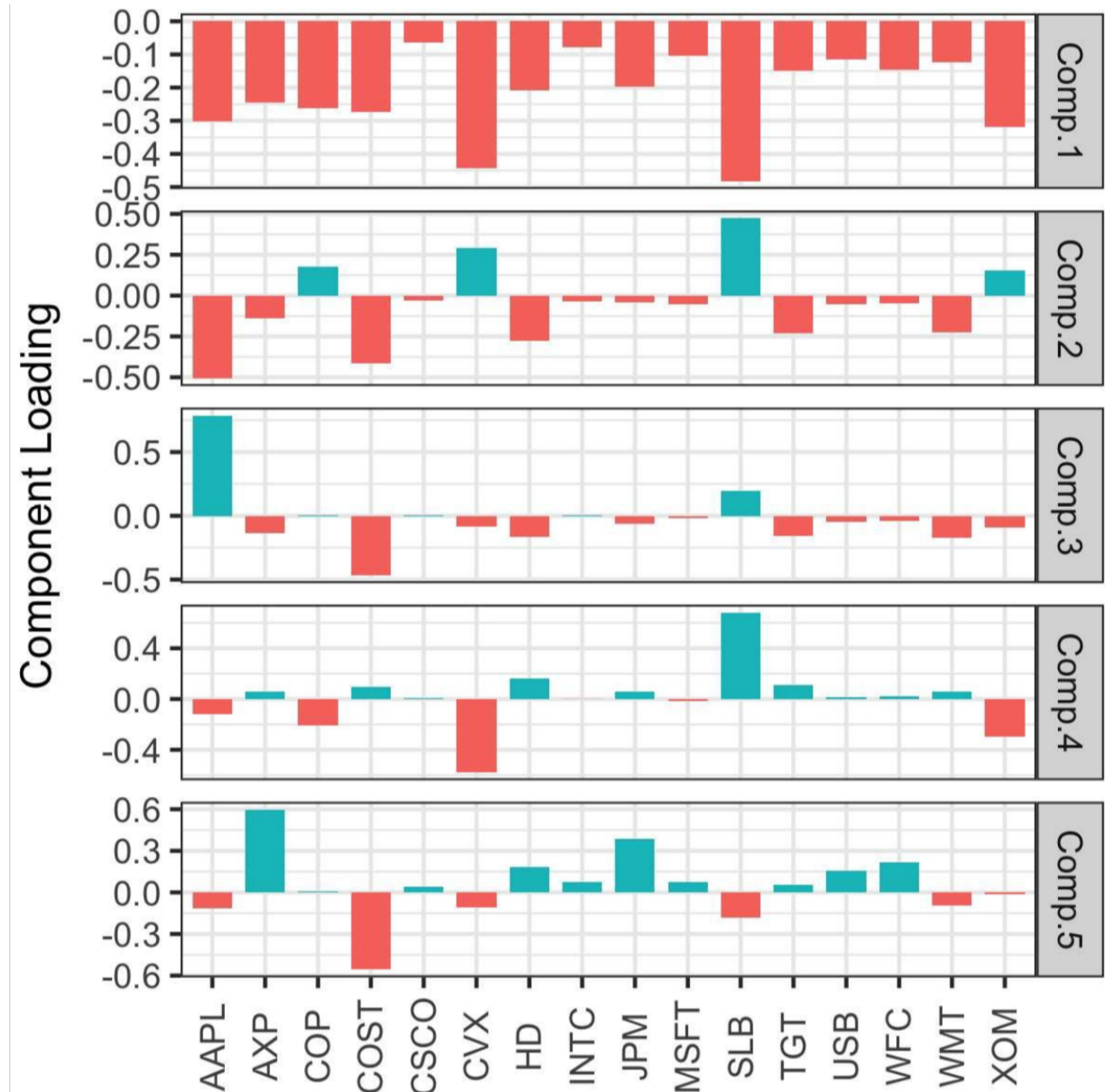


Figure 7-3. The loadings for the top five principal components of stock price returns



<https://forms.gle/HAc9XiVoV1gsAMZH8>

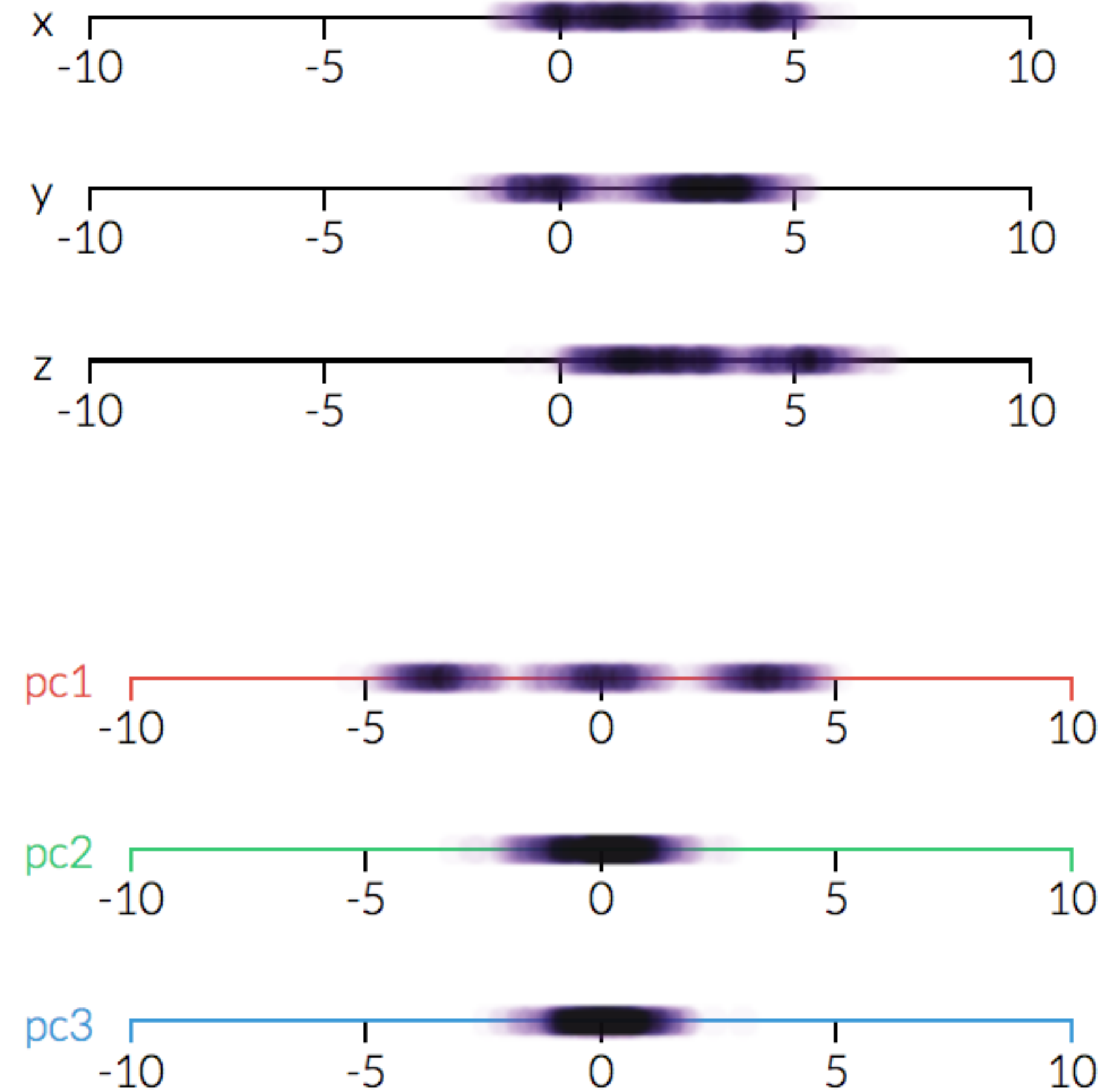
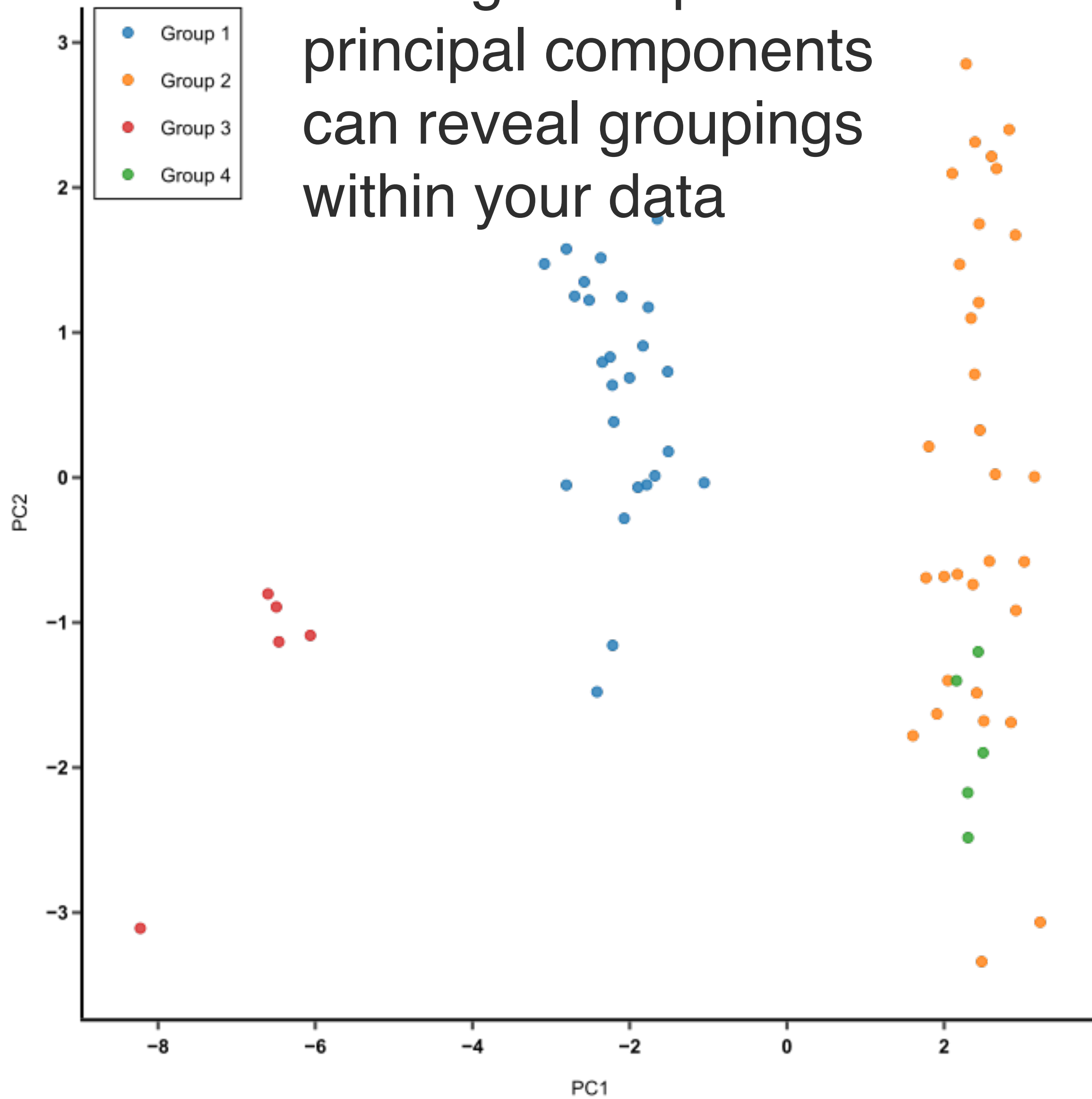
For more on PCA:

- <https://blog.bioturing.com/2018/06/14/principal-component-analysis-explained-simply/>
- <http://setosa.io/ev/principal-component-analysis/>

PCA : Key Ideas

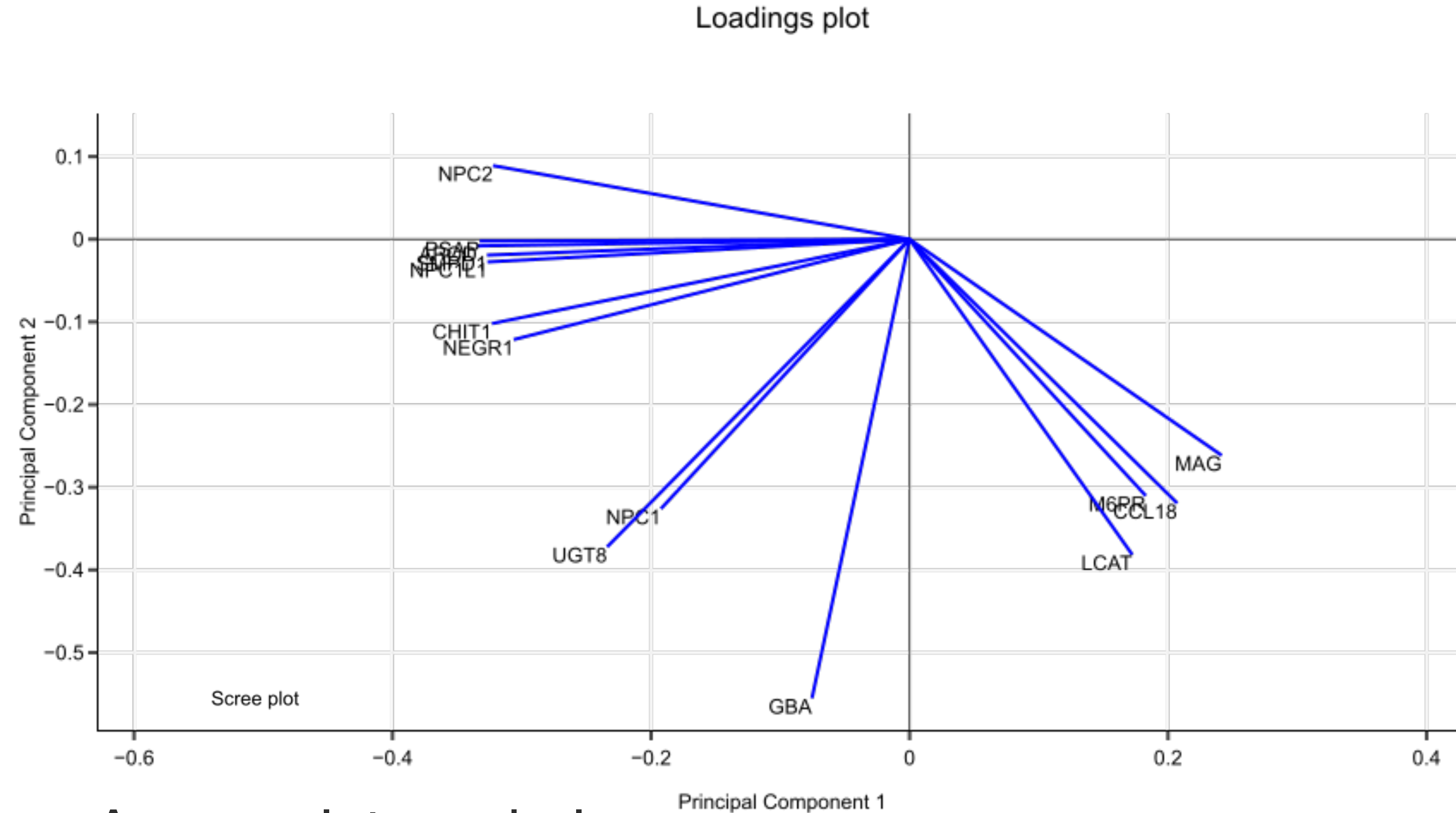
1. PCs are linear combinations of the predictor variables (numeric data only)
2. Calculated to minimize correlation between components (minimizes redundancy)
3. A limited number of components will typically explain most of the variance in the outcome variable
4. Limited set of PCs can be used in place of original predictors (dimensionality reduction)

Plotting the top principal components can reveal groupings within your data

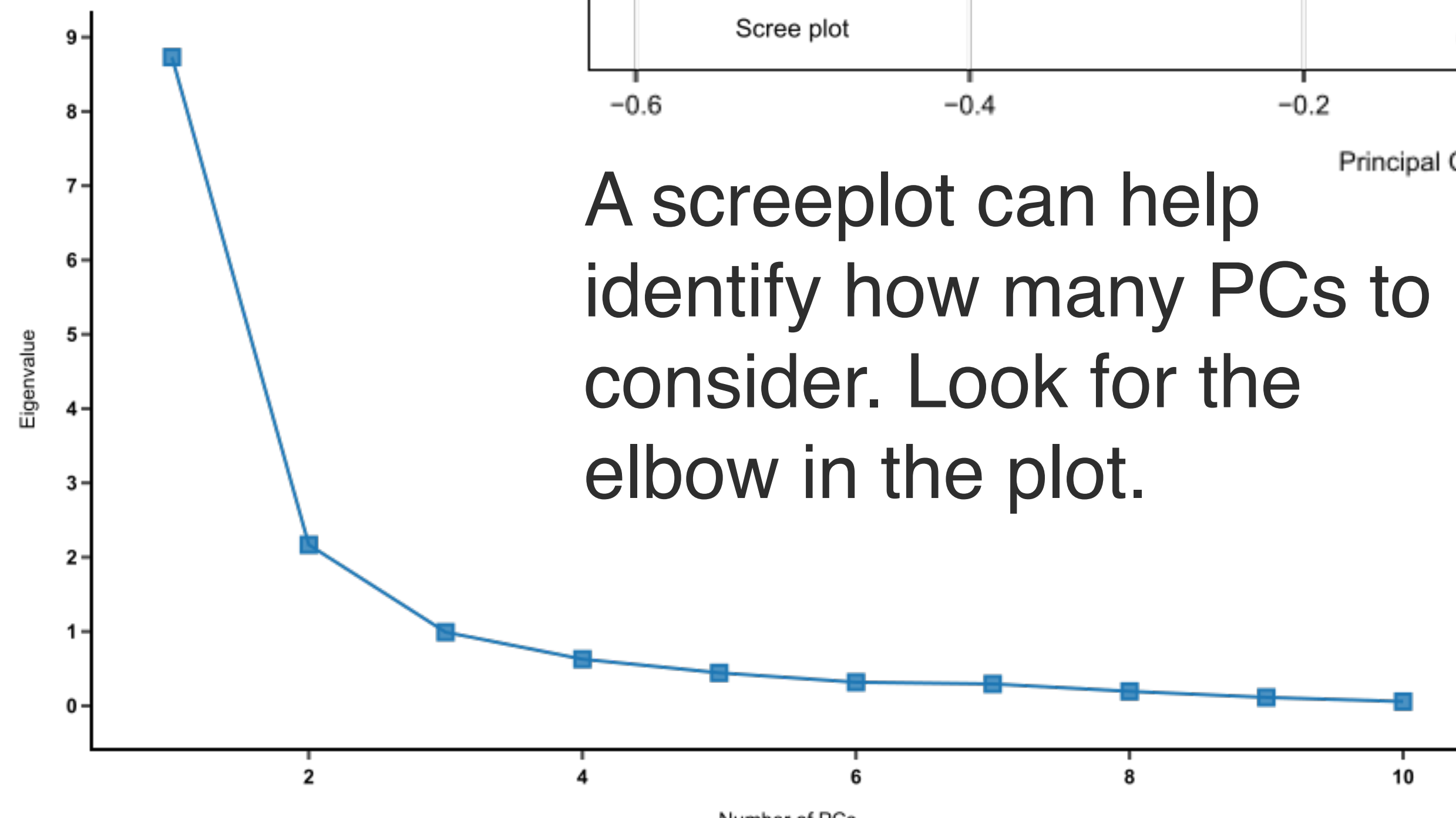


Loadings plot: project values on each PC show how much weight they have on that PC

In this example, NPC2 and CHIT1 strongly influence PC1, while GBA and LCAT have more say in PC2



A screeplot can help identify how many PCs to consider. Look for the elbow in the plot.

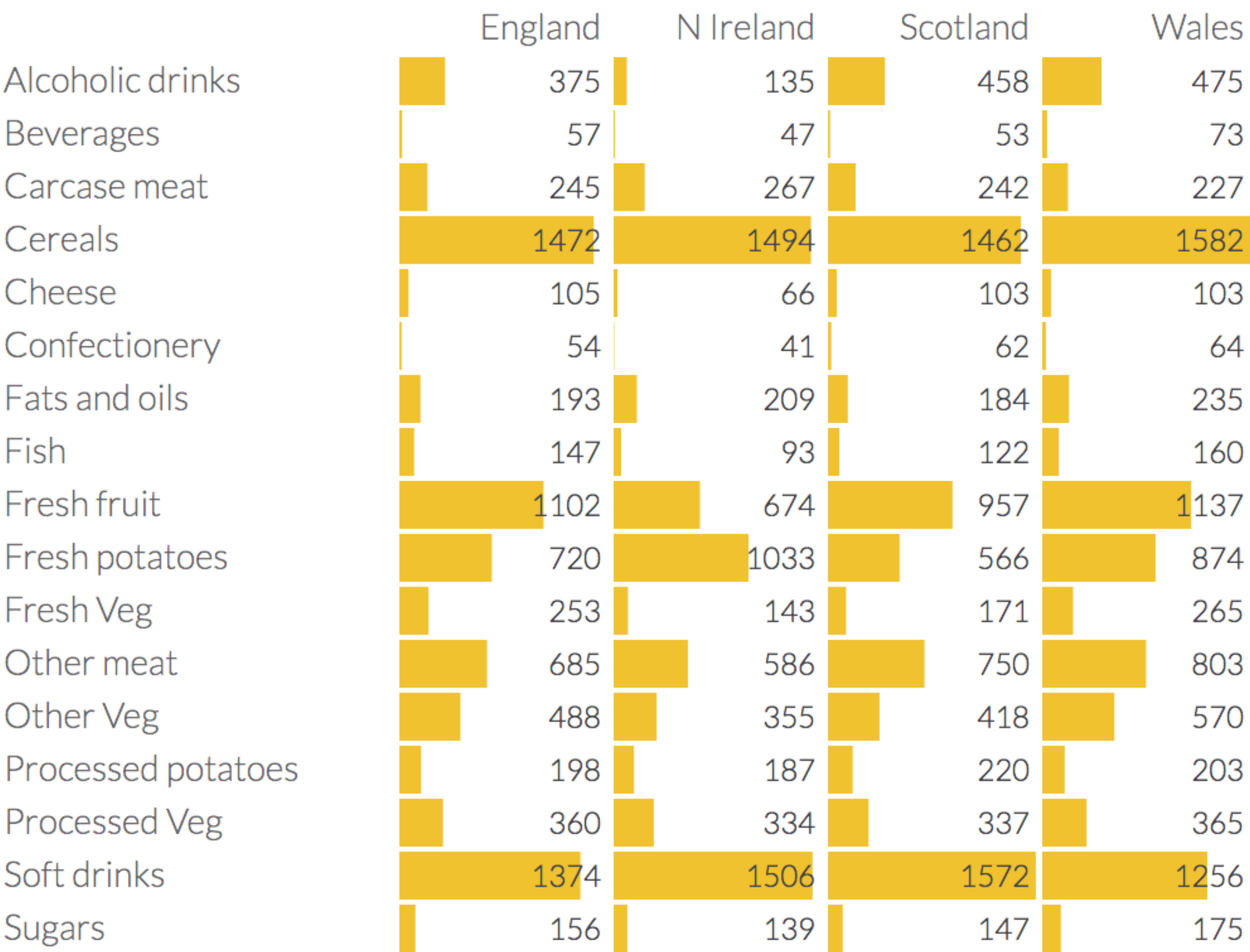


Examples

```
>>> import numpy as np
>>> from sklearn.decomposition import PCA
>>> X = np.array([[-1, -1], [-2, -1], [-3, -2], [1, 1], [2, 1], [3, 2]])
>>> pca = PCA(n_components=2)
>>> pca.fit(X)
PCA(n_components=2)
>>> print(pca.explained_variance_ratio_)
[0.9924 0.0075]
>>> print(pca.singular_values_)
[6.30061 0.54980]
```

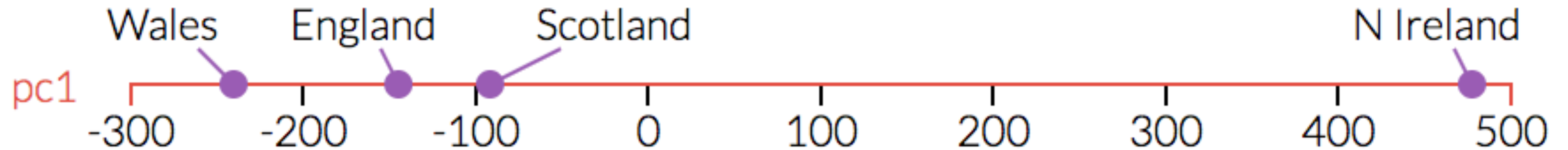
```
>>> pca = PCA(n_components=2, svd_solver='full')
>>> pca.fit(X)
PCA(n_components=2, svd_solver='full')
>>> print(pca.explained_variance_ratio_)
[0.9924 0.00755]
>>> print(pca.singular_values_)
[6.30061 0.54980]
```

Case Study: Diet in the UK



1 / 1000s x 4 countries

PCA: Diet in the K



Can already tell that N Ireland is distinct

If we look back at the raw data
Northern Ireland, eats way more
fresh potatoes and way fewer fresh
fruits, cheese, & fish. This reflects
real world geography...



Case Study: Genetics and Geography

Letter | Published: 31 August 2008

Genes mirror geography within Europe

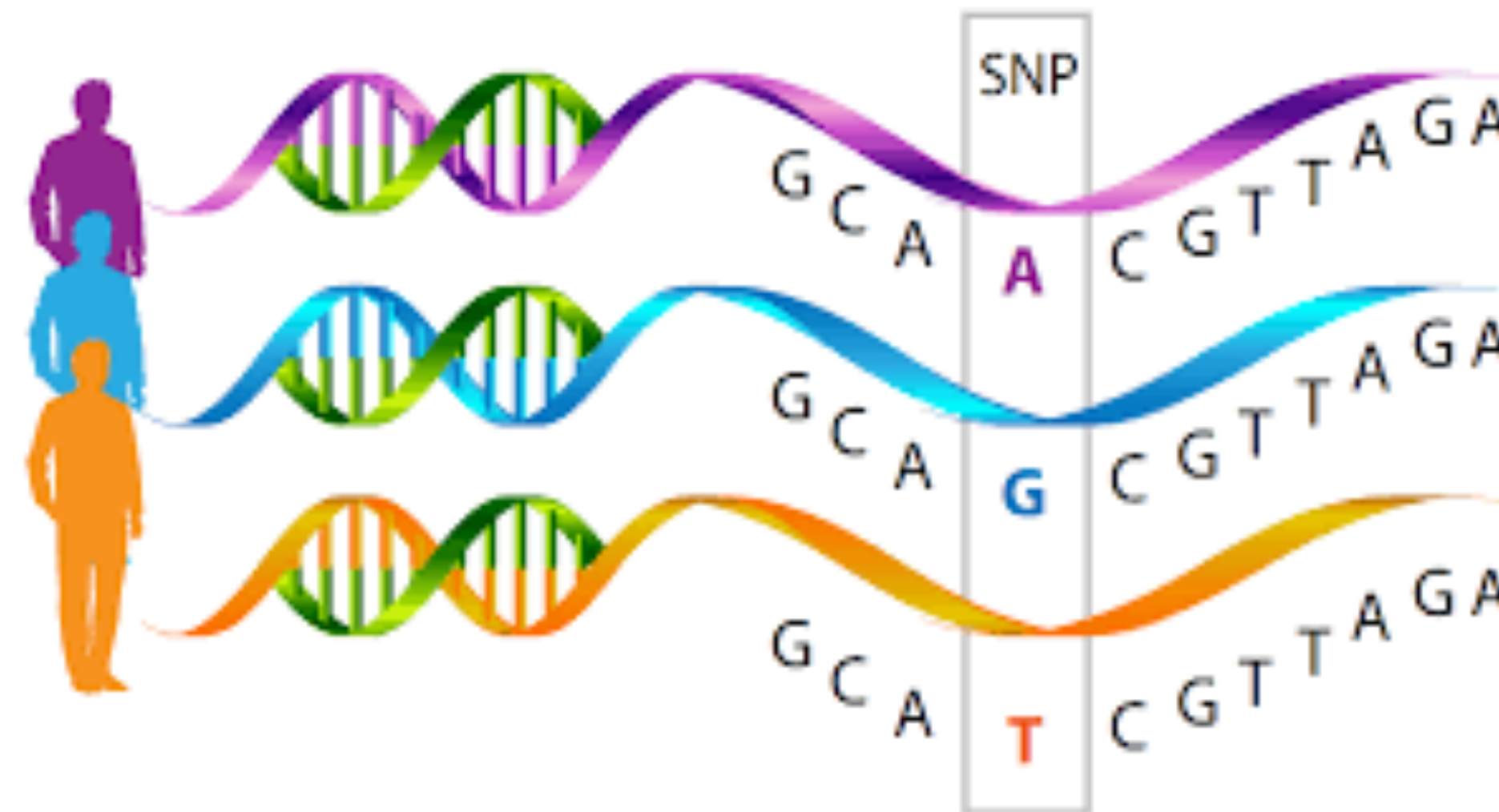
John Novembre , Toby Johnson, Katarzyna Bryc, Zoltán Kutalik, Adam R. Boyko, Adam Auton, Amit Indap, Karen S. King, Sven Bergmann, Matthew R. Nelson, Matthew Stephens & Carlos D. Bustamante

Nature **456**, 98–101 (06 November 2008) | [Download Citation](#) 

The Data: 1,387 Europeans x 500,000 SNPs

SNP (Single Nucleotide Polymorphism)

- Reminder: Your DNA is made up of four bases: G, C, T, & A
- A SNP is a position in one's DNA that varies between individuals (appears in at least 1% of the population)
 - This results from normal human variation
 - Some contribute to disease, but many are just differences between humans
 - These are used by companies like 23andMe and Ancestry.com



The Data: 1,387 Europeans x 500,000 SNPs

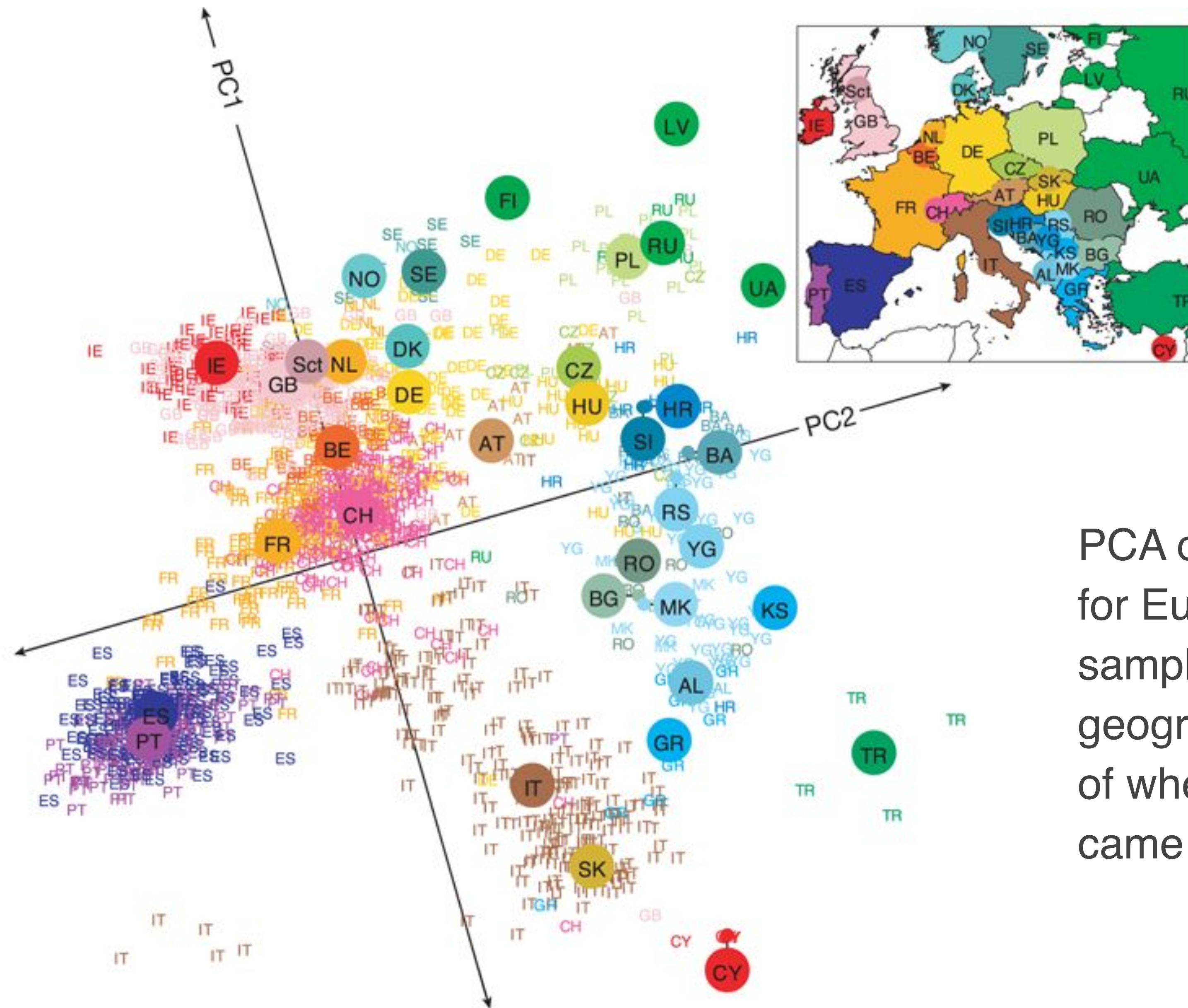
Step 1: Measure genotype (GCTA) at 500,000 positions (SNPs) along the genome in 1387 European individuals

Step 2: Calculate PCs from 500,000 SNPs

Step 3: Plot PC1 and PC2 (each point is an individual)

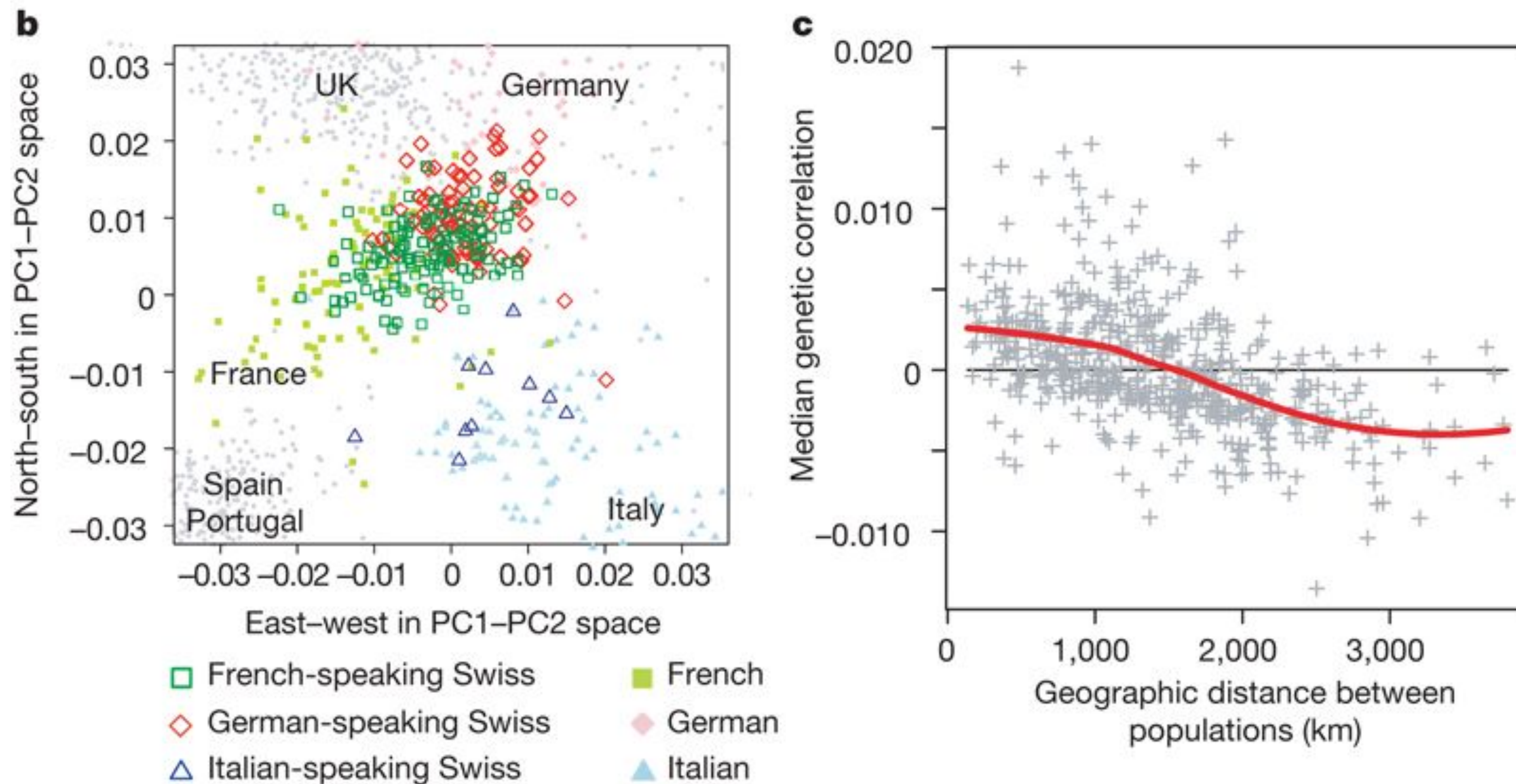
Step 4: Compare to the map of Europe

a



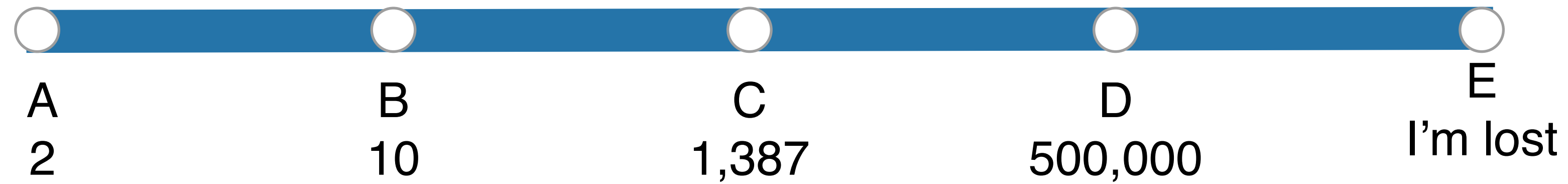
PCA on SNP data for European samples reflects geographic location of where samples came from

PC1 is East-West ; PC2 is North-South



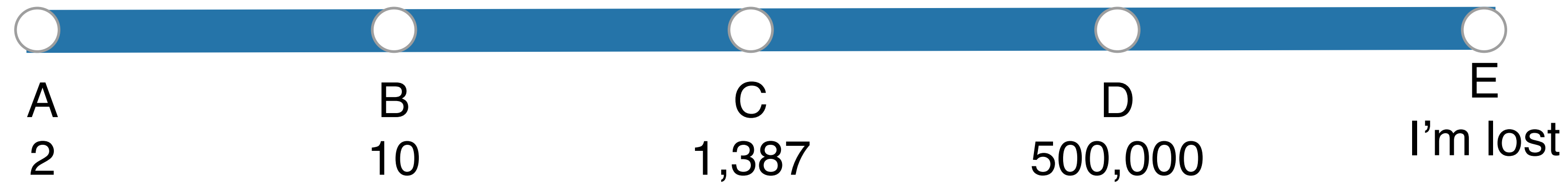


This analysis used 500,000 SNPs from 1,387 individuals.
How many PCs would have been calculated?





This analysis used 500,000 SNPs from 1387 individuals. How many PCs explain geographic differences across Europe by genetic ancestry?



- 
- 
- A** PC1 explains geographic differences from North to South
- B** PC2 explains geographic differences from East to West
- C** The French (FR) are not genetically related to the Scottish (Sct)
- D** The French are more closely related genetically to Germans (DE) than they are the Fins (GL)
- E** The Spanish (ES) and Portuguese (PT) are genetically similar

Dimensionality Reduction with PCA: Pros & Cons

Pros:

- Helps compress data; reduced storage space.
- reduces computation time.
- helps remove redundant features (if any)
- Identifies outliers in the data

Cons:

- may lead to some amount of data loss.
- finds linear relationships between variables, there are times that's not enough.
- fails in cases where mean and covariance are not enough to define datasets.
- may not know how many principal components to keep
- highly affected by outliers in the data

Manifold learning

PCA is linear, this is not

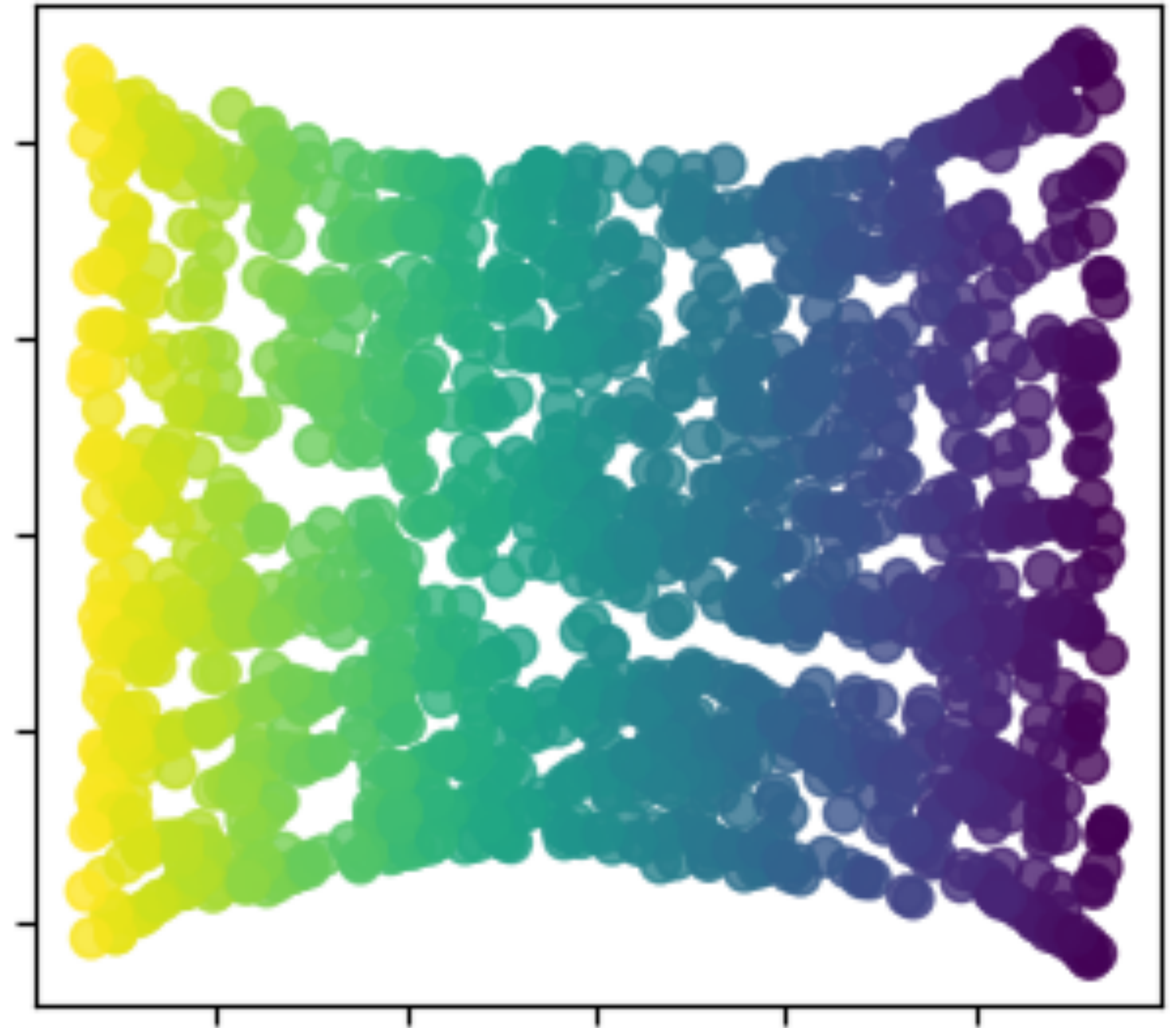
- Manifold learning is an approach to dimensionality reduction where the problem is only locally linear/Euclidean
- Algorithms for this task are based on the idea that the dimensionality of many data sets is only artificially high if you allow curved manifolds



Manifold learning

- Manifold learning is an approach to dimensionality reduction where the problem is only locally linear/Euclidean
- Algorithms for this task are based on the idea that the dimensionality of many data sets is only artificially high if you allow curved manifolds

“Idealized” 2D manifold for Swiss Roll

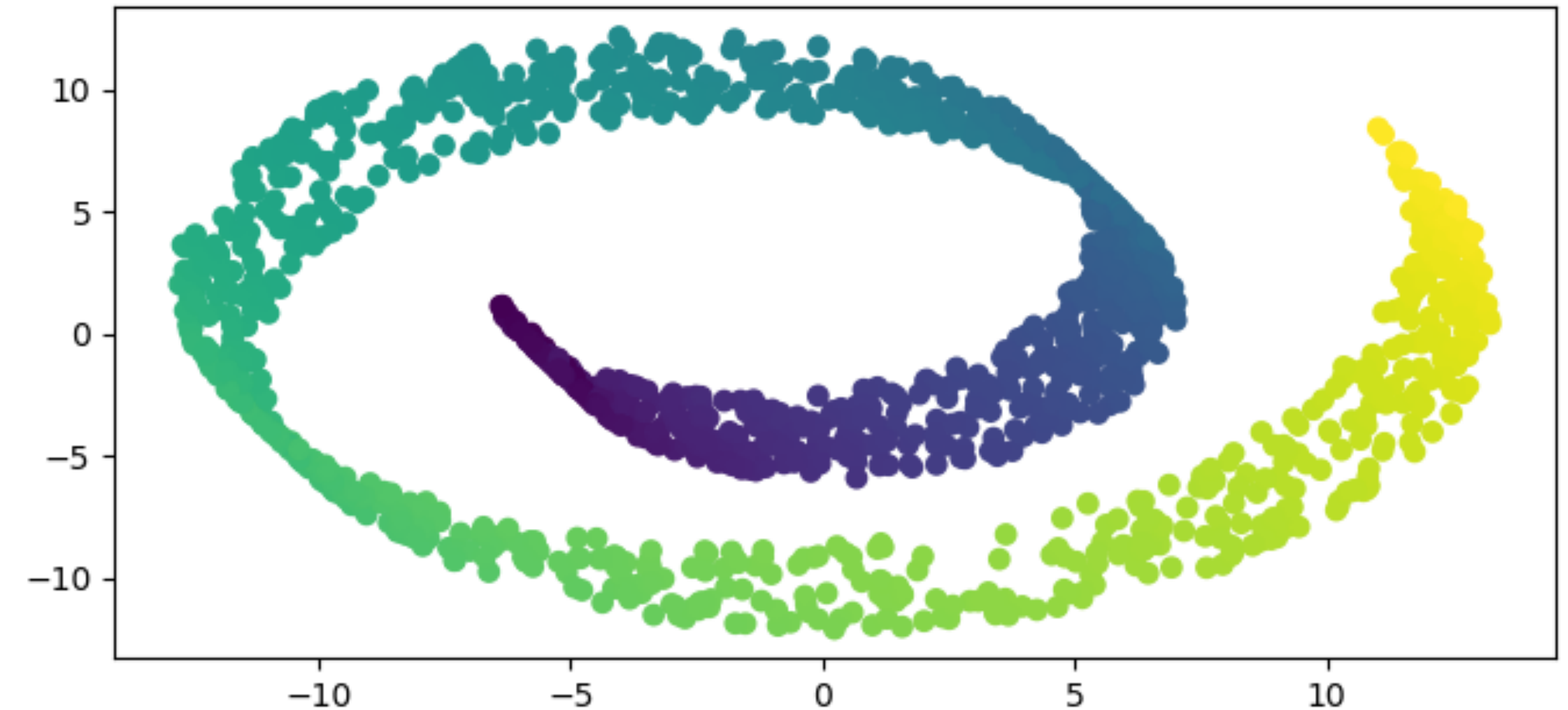


t-SNE

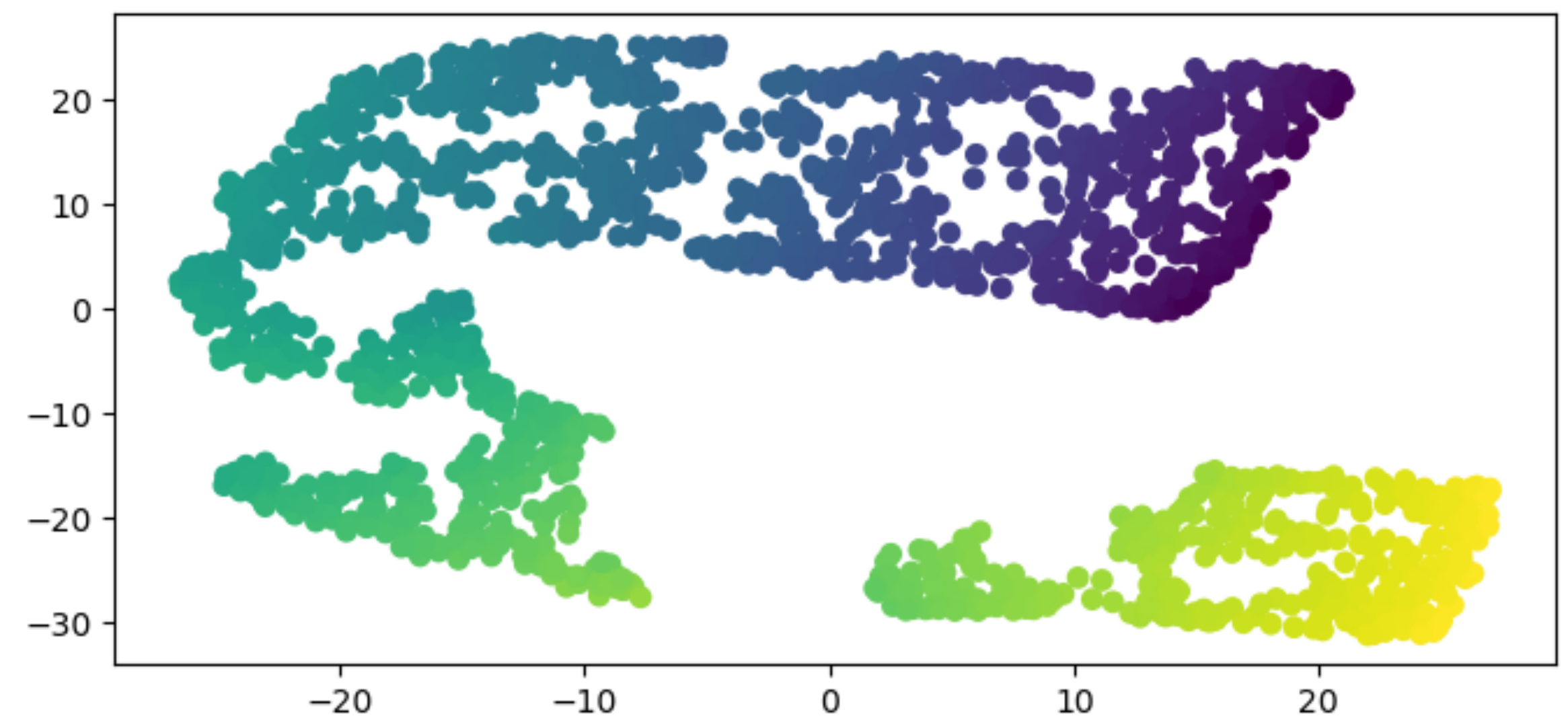
t-distributed Stochastic Neighbor Embedding

- PCA tries to find a global structure
 - Can lead to local inconsistencies in a subspace...far away points can become nearest neighbors
- t-SNE tries to preserve local structure
 - “For high-dimensional data that lies on or near a low-dimensional, non-linear manifold it is usually more important to keep the low-dimensional representations of very similar datapoints close together, which is typically not possible with a linear mapping”

PCA of Swiss Roll Dataset



t-SNE of Swiss Roll Dataset

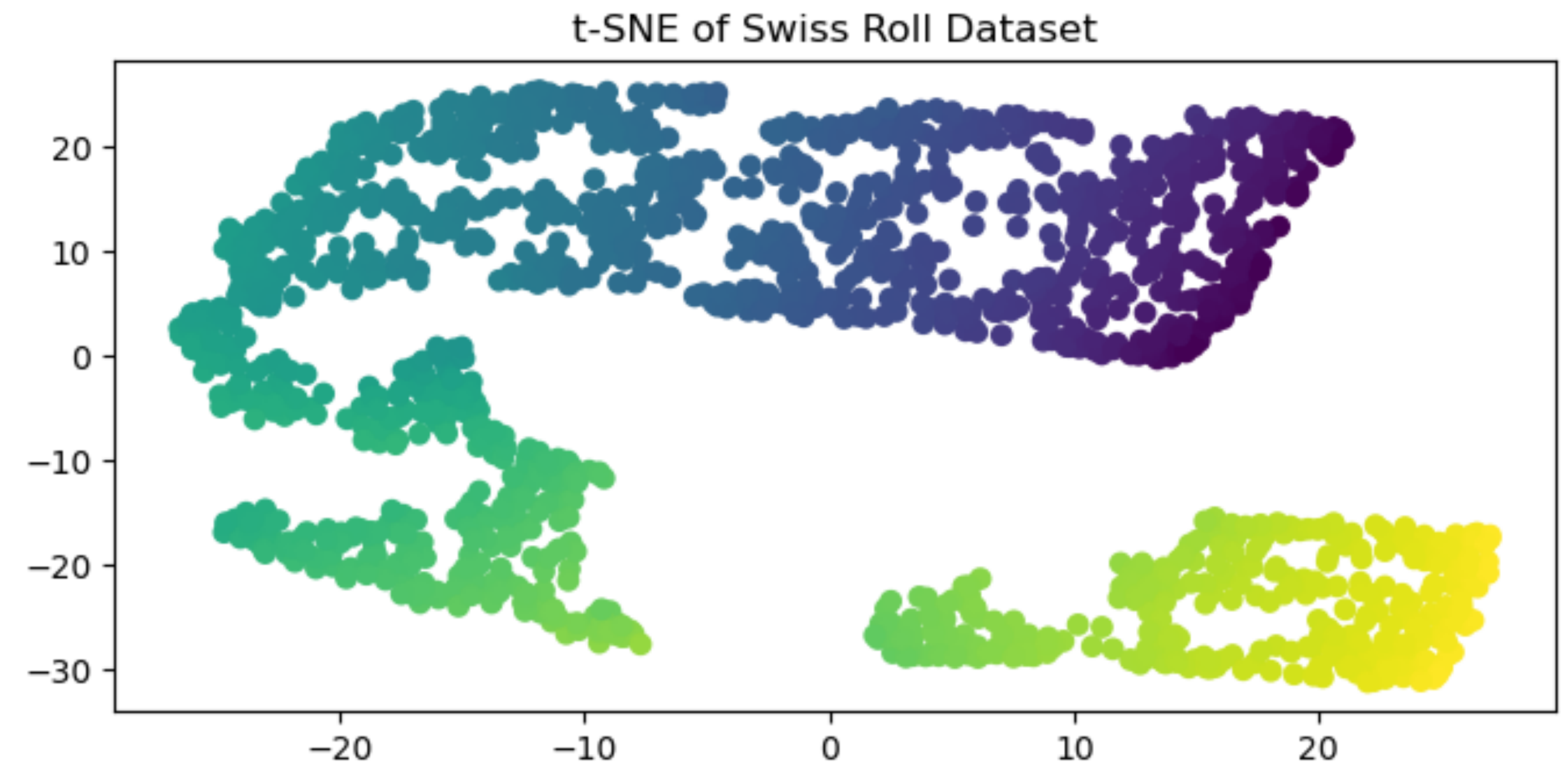


t-SNE

t-distributed Stochastic Neighbor Embedding

- t-SNE has a critical parameter you can choose!
- “Perplexity” explicitly trades off how much we care about global structure vs local structure
- It’s not this, but you can think of perplexity as the number of “close neighbors” a data point should have on average
- Lower perplexity -> local structure
- Higher perplexity -> global structure

```
t_sne = manifold.TSNE(  
    n_components=2,  
    perplexity=30,  
    init="random",  
    max_iter=250,  
    random_state=0,  
)  
S_t_sne = t_sne.fit_transform(S_points)
```



<https://distill.pub/2016/misread-tsne/>

How to Use t-SNE Effectively

Although extremely useful for visualizing high-dimensional data, t-SNE plots can sometimes be mysterious or misleading. By exploring how it behaves in simple cases, we can learn to use it more effectively.



Step
44

Number Of Points 50



Perplexity 10



Epsilon 5



Points arranged in 3D,
on two linked circles.
Different runs may give
different results.

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